MATHEMATICS (Senior Secondary) – PRACTICE TEST

Module IV: Co-ordinate Geometry (Lessons 13 – 16)

Cartesian System of Rectangular Co-ordinates · Straight Lines · Circles · Conic Sections

Time: 2 hours	**Maximum Marks:** 50
Name: _____	**Date:** _____

General Instructions: *(i) All questions are compulsory. (ii) Marks are shown against each question. (iii) Show all steps of your working. (iv) Use of a calculator is not permitted.*

SECTION A – Multiple Choice Questions (8 × 1 = 8 marks)

Choose the correct option.

1. The distance between the points (3, –2) and (–1, 1) is

(a) 3 units (b) 4 units (c) 5 units (d) 7 units

2. The slope of the line $5x - 2y + 7 = 0$ is

(a) –5/2 (b) 2/5 (c) –2/5 (d) 5/2

3. The centre of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ is

(a) (3, –2) (b) (–3, 2) (c) (6, –4) (d) (–6, 4)

4. The focus of the parabola $y^2 = 12x$ is

(a) (12, 0) (b) (0, 3) (c) (3, 0) (d) (6, 0)

5. The eccentricity of the ellipse $x^2/25 + y^2/16 = 1$ is

(a) 3/5 (b) 4/5 (c) 3/4 (d) 5/3

6. When the origin is shifted to the point (2, –3) by translation of axes, the new co-ordinates of the point (5, 1) are

(a) (7, –2) (b) (3, 4) (c) (–3, –4) (d) (3, –2)

7. The length of the transverse axis of the hyperbola $x^2/9 - y^2/16 = 1$ is

(a) 3 units (b) 4 units (c) 8 units (d) 6 units

8. The y-intercept of the line $3x - 4y + 12 = 0$ is

(a) 3 (b) –3 (c) 4 (d) –4

SECTION B – Short Answer Questions I (6 × 2 = 12 marks)

9. Find the co-ordinates of the point which divides the line segment joining A(2, –1) and B(7, 9) internally in the ratio 2 : 3.

10. Show that the points A(–1, 2), B(2, 8) and C(4, 12) are collinear.

11. Find the equation of the line passing through the point (3, –4) and having slope –2/3.

12. Find the perpendicular distance of the point (1, –2) from the line $4x + 3y - 10 = 0$.

13. Find the equation of the circle with centre (–2, 3) and radius 5 units. Write it in the form $x^2 + y^2 + 2gx + 2fy + c = 0$.

14. Find the equation of the parabola whose vertex is at the origin and whose focus is (0, –3).

SECTION C – Short Answer Questions II (5 × 4 = 20 marks)

15. The vertices of a triangle ABC are A(2, 3), B(–1, 0) and C(4, –2). Find (a) the area of the triangle ABC and (b) the length of the altitude drawn from A to the side BC.

16. The mid-points of the sides BC, CA and AB of a triangle ABC are D(2, 1), E(3, 4) and F(–1, 2) respectively. Find the co-ordinates of the vertices A, B, C and of the centroid of the triangle.

17. Find the equation of the straight line which passes through the point of intersection of the lines $x + 2y - 5 = 0$ and $3x - y - 1 = 0$ and also passes through the point (2, 3).

18. Find the equation of the circle passing through the points (1, 2), (5, 2) and (1, –2). Also write down its centre and radius.

19. Find the equation of the ellipse (referred to its centre) whose foci are (±4, 0) and whose eccentricity is 4/5. Also find the length of its latus rectum.

SECTION D – Long Answer Questions (2 × 5 = 10 marks)

20. Consider the lines $3x + 4y - 7 = 0$ and $6x + 8y + 16 = 0$.

(a) Show that the two lines are parallel. [1]

(b) Find the distance between them. [2]

(c) Find the equation of the line which is parallel to both and lies midway between them. [2]

21. For the hyperbola $16x^2 - 9y^2 = 144$, find (i) the eccentricity, (ii) the co-ordinates of the foci, (iii) the co-ordinates of the vertices, (iv) the equations of the directrices and (v) the length of the latus rectum. [1 each]

— End of Question Paper —

ANSWER KEY AND MARKING SCHEME

SECTION A

Q.	1	2	3	4	5	6	7	8
Ans.	(c) 5	(d) 5/2	(a) (3, –2)	(c) (3, 0)	(a) 3/5	(b) (3, 4)	(d) 6	(a) 3

Hints: Q1 √(16+9)=5; Q2 m = –A/B = 5/2; Q3 (–g, –f); Q4 4a = 12 ⇒ a = 3; Q5 e^2 = 1 – 16/25 = 9/25; Q6 x′ = 5–2, y′ = 1+3; Q7 2a = 6; Q8 put x = 0 ⇒ y = 3.

SECTION B

9. x = (2·7 + 3·2)/(2+3) = 20/5 = 4; y = (2·9 + 3·(–1))/5 = 15/5 = 3. **Point (4, 3)** [1 + 1]

10. Area = ½ |(–1)(8–12) + 2(12–2) + 4(2–8)| = ½ |4 + 20 – 24| = 0 [1½]. Area is zero, so the points are collinear [½]. (Equivalent: slope AB = slope BC = 2.)

11. $y - y_1 = m(x - x_1) \Rightarrow y + 4 = (-2/3)(x - 3) \Rightarrow 3y + 12 = -2x + 6 \Rightarrow$ **2x + 3y + 6 = 0** [1 + 1]

12. d = |4(1) + 3(–2) – 10| / √(4²+3²) = |–12|/5 = **12/5 units** [1 + 1]

13. $(x + 2)^2 + (y - 3)^2 = 25$ [1] ⇒ $\mathbf{x^2 + y^2 + 4x - 6y - 12 = 0}$ [1]

14. Focus (0, –a) with a = 3 ⇒ standard form $x^2 = -4ay$ [1] ⇒ $\mathbf{x^2 = -12y}$ [1]

SECTION C

15. (a) Area = ½ |2(0+2) + (–1)(–2–3) + 4(3–0)| = ½ |4 + 5 + 12| = **21/2 sq. units** [2]

(b) BC = √((4+1)² + (–2–0)²) = √29 [1]. Area = ½ × BC × h ⇒ h = 2 × (21/2)/√29 = **21/√29 units** [1]

16. If $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$: D, E, F are mid-points of BC, CA, AB, so A = E + F – D, B = D + F – E, C = D + E – F. [1]

A = (3 – 1 – 2, 4 + 2 – 1) = **(0, 5)**; B = (2 – 1 – 3, 1 + 2 – 4) = **(–2, –1)**; C = (2 + 3 + 1, 1 + 4 – 2) = **(6, 3)** [2]

Centroid = ((0–2+6)/3, (5–1+3)/3) = **(4/3, 7/3)** [1]

17. Family: (x + 2y – 5) + λ(3x – y – 1) = 0 [1]. Through (2, 3): (2 + 6 – 5) + λ(6 – 3 – 1) = 0 ⇒ 3 + 2λ = 0 ⇒ λ = –3/2 [1½].

Substituting: 2(x + 2y – 5) – 3(3x – y – 1) = 0 ⇒ –7x + 7y – 7 = 0 ⇒ **x – y + 1 = 0** [1½]

18. Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$. Using the three points: 2g + 4f + c = –5, 10g + 4f + c = –29, 2g – 4f + c = –5. [1½]

Subtracting: g = –3, f = 0, c = 1 [1½]. Equation: $\mathbf{x^2 + y^2 - 6x + 1 = 0}$. Centre (3, 0), radius = √(9 + 0 – 1) = **2√2** [1]

19. Foci (±ae, 0) with ae = 4 and e = 4/5 ⇒ a = 5 [1]. $b^2 = a^2(1 - e^2) = 25 - 16 = 9$ [1]. Equation: $\mathbf{x^2/25 + y^2/9 = 1}$ [1]. Latus rectum = $2b^2/a$ = **18/5** [1]

SECTION D

20. (a) Dividing the second line by 2: 3x + 4y + 8 = 0. Coefficients of x and y are proportional (3 : 4 in both) while the constants differ, so the lines are parallel. [1]

(b) Distance = $|C_1 - C_2|/\sqrt{(A^2+B^2)}$ = |–7 – 8|/√(9+16) = 15/5 = **3 units** [2]

(c) The required line is 3x + 4y + k = 0 with k = (–7 + 8)/2 = 1/2, i.e. **6x + 8y + 1 = 0**. (Check: its distance from each given line is 3/2.) [2]

21. Divide by 144: $x^2/9 - y^2/16 = 1$, so $a^2 = 9$, $b^2 = 16$ ($a = 3$, $b = 4$). [1 for the form]

(i) $e = \sqrt{((a^2+b^2)/a^2)} = \sqrt{(25/9)} =$ **5/3**

(ii) Foci $(\pm ae, 0) =$ **$(\pm 5, 0)$**

(iii) Vertices $(\pm a, 0) =$ **$(\pm 3, 0)$**

(iv) Directrices $x = \pm a/e = \pm 3/(5/3) \Rightarrow$ **$x = \pm 9/5$**

(v) Latus rectum $= 2b^2/a = 2 \cdot 16/3 =$ **32/3**

(1 mark each for (i)–(v); the working step for the standard form is part of (i).)